beyond our present capacity, each would be seen to be made of a single, tiny, oscillating loop of string.

For reasons that will become Clear, the length of a typical string loop is about the Planck length, about a hundred billion billion (1020) times smaller than an atomic nucleus. It is no wonder that our present-day experiments are unable to resolve the microscopic stringy nature of matter: strings are minute even on the scales set by subatomic particles. We would need an accelerator to slam matter together with energies some million billion times more powerful than any previously constructed in order to reveal directly that a string is not a point-particle.

We will describe shortly the stunning implications that follow from replacing point-particles by strings, but let's first address a more basic question: What are strings made of?

There are two possible answers to this question. First, strings are truly fundamental—they are "atoms," uncuttable constituents, in the truest sense of the ancient Greeks. As the absolute smallest constituents of anything and everything, they represent the end of the line—the last of the Russian matrioshka dolls—in the numerous layers of substructure in the microscopic world. From this perspective, even though strings have spatial extent, the question of their composition is without any content. Were strings to be made of something smaller they would not be fundamental. Instead, whatever strings were composed of would immediately displace them and lay claim to being an even more basic constituent of the universe. Using our linguistic analogy, paragraphs are made of sentences, sentences are made of words, and words are made of letters. What makes up a letter? From a linguistic standpoint, that's the end of the line. Letters are letters—they are the fundamental building blocks of written language; there is no further substructure. Questioning their composition has no meaning. Similarly, a string is simply a string—as there is nothing more fundamental, it can't be described as being composed of any other substance. That's the first answer. The second answer is based on the simple fact that as yet we do not know if string theory is a correct or final theory of nature. If string theory is truly off the mark, then, well, we can forget strings and the irrelevant question of their composition. Although this is a possibility, research since the mid-1980s